



আন্তর্জাতিক ইসলামী বিশ্ববিদ্যালয় চট্টগ্রাম
الجامعة الإسلامية العالمية سبّاغونغ
International Islamic University Chittagong

LAB REPORT

Experiment No. : 01
Experiment Name : Familiarization with the necessary resources of Digital Electronic Sessional.
Course Title : Digital Logic Design Lab.
Course Code : CSE-2324
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Semester : 3rd
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Date of Submission : 09-09-2026
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Adjunct lecturer, Dept. of CSE, IIUC

REMARK

Experiment name: Familiarization with the necessary resources of Digital Electronics Sessional.

Objective:

1. To identify and understand the working principles of essential digital laboratory instruments and tools.
2. To study the pin configuration and datasheets of common logic gate IC's (7408, 7432, 7404, 7400, 7402).
3. To learn how to construct and verify basic logic circuits on a trainer kit using switches and LED indicators.

Apparatus:

- 1) Digital electronics trainer kit.
- 2) DC power supply.
- 3) Connecting wires.
- 4) Digital IC's:
 - IC 7408: AND Gate.
 - IC 7432: OR Gate.
 - IC 7404: NOT Gate.
 - IC 7400: NAND Gate.
 - IC 7402: NOR Gate.

Truth Table:

a) Name of the IC: IC 7408, AND Gate:

A	B	$X=A.B$
0	1	0
1	0	0
1	1	1
0	0	0

b) Name of the IC: IC 7432, OR Gate:

A	B	$X=A+B$
0	1	1
1	0	1
1	1	1
0	0	0

c) Name of the IC: IC 7404, NOT Gate:

A	$X = \bar{A}$
0	1
1	0

d) Name of the IC: IC 7400, NAND Gate:

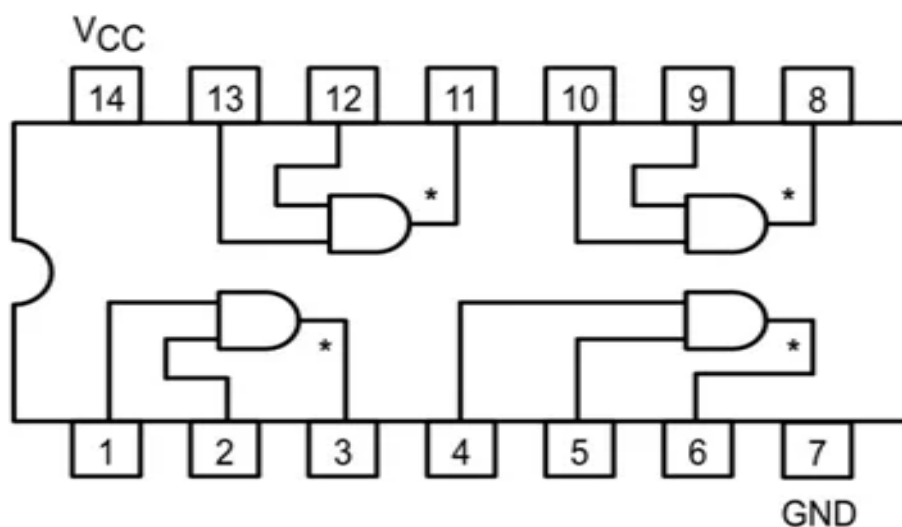
A	B	$X = \overline{A \cdot B}$
0	1	1
1	0	1
1	1	0
0	0	1

e) Name of the IC: IC 7402, NOR Gate:

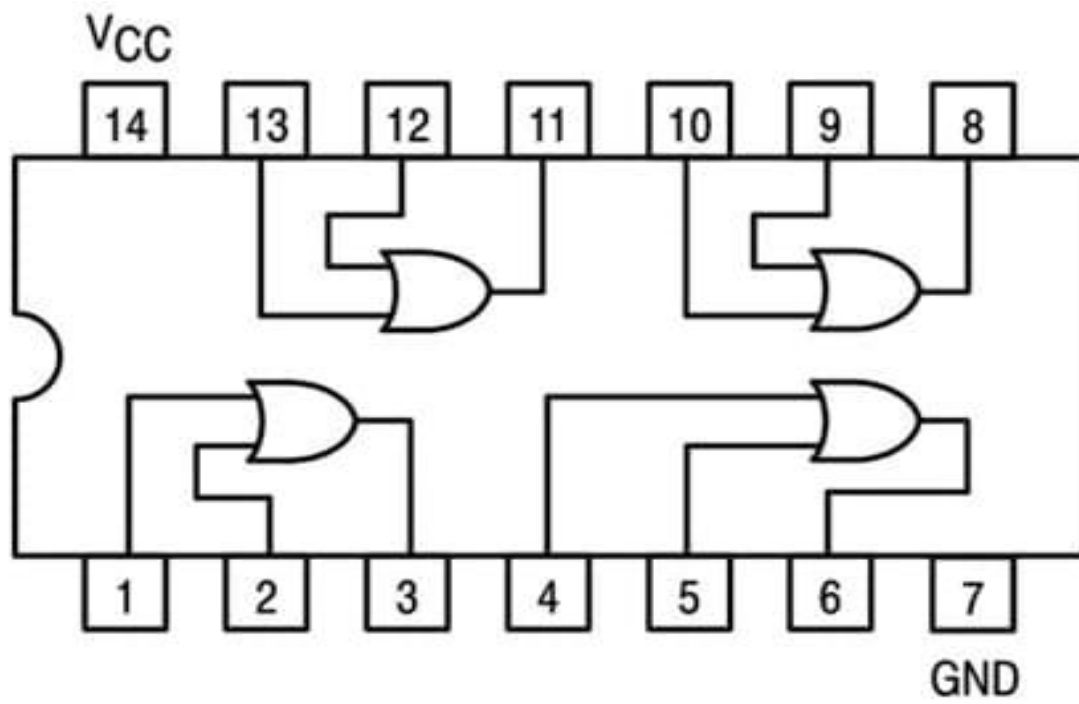
A	B	$X = \overline{A + B}$
0	1	0
1	0	0
1	1	0
0	0	1

Pin Diagram:

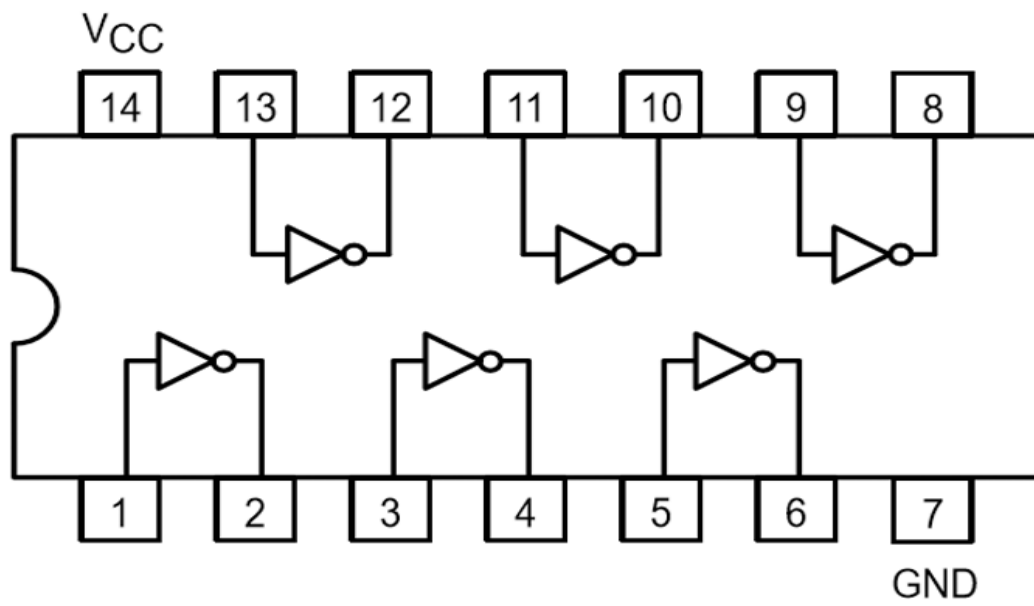
a) IC 7408, AND Gate:



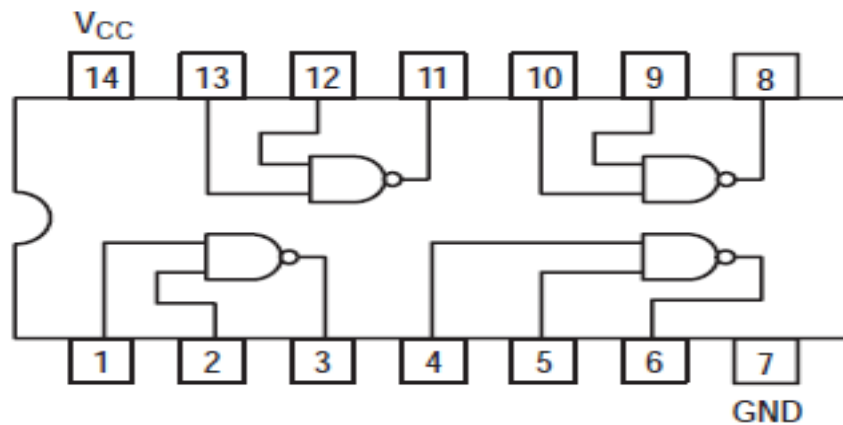
b) IC 7432, OR Gate:



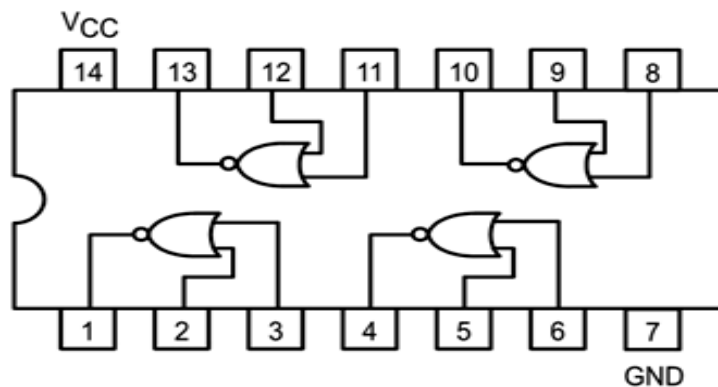
c) IC 7404, NOT Gate:



d) IC 7400, NAND Gate:



e) IC 7402, NOR Gate:



Conclusion:

In this experiment, the logic functions and pin configurations of basic and universal logic gates: AND (IC 7408), OR (IC 7432), NOT (IC 7404), NAND (IC 7400), and NOR (IC 7402)—were successfully verified using the digital electronics trainer kit. All experimental outputs matched their theoretical truth tables, with proper power connections ($V_{CC} = +5V$ and GND) maintained throughout. Additionally, the non-standard pin configuration of IC 7402 (NOR Gate) was accurately identified and tested.